

Trade Theory: Manufacturing Process Control and Automation

Government ITI Trades Portal

Category: Emerging Technologies

Duration: 1 year

Eligibility: 10th class pass

Description

MPCA teaches students how to run and control modern factories using computers, sensors, and automatic machines. Students learn to operate PLC, SCADA, robots, and sensors used in large factories.

Core Syllabus and Curriculum

Core syllabus and curriculum

- Basics: electrical, electronics, sensors, motors
- PLC: Programmable Logic Controller
- Pneumatics and hydraulics: air and oil pressure machines
- HMI: touch screen panels used by operators
- Project: automate a small bottling / packaging line

Apprenticeship Opportunities

Government Companies

- DRDO Defence Research and Development Organisation
- BHEL Bharat Heavy Electricals Limited
- Railways
- Aerospace engineering

Private Companies

- Tata Motors
- Maruti Suzuki Ltd
- Honda
- Nestle

Job Opportunities

Government Jobs

- Railways

Private Sector

- Automation engineering
- Manufacturing engineering
- Quality engineering

Salary Range

Fresher: Rs. 20,000-35,000/month; Experienced/licensed: Rs. 60,000-80,000/month